



Usability Testing For Park At Ease Part-8B

25/VS_INFO_3201_V1

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Usability Test Part 2: Participant Transcript Analysis

To complement the structured usability testing conducted earlier, we further analyzed the qualitative feedback from **five participants** who interacted with our prototype of the *Park At Ease* application. These user sessions were guided using a predefined scenario and conducted in a controlled setting with direct observation, voice recording, and post-test interviews.

Participant 1

- **Experience:** Found the app “pretty straightforward” and was able to complete the task in about 30 seconds without any help.
- **Positive Feedback:** Liked how easy it was to locate parking using filters; praised the efficiency of the location-based filtering.
- **Issues Noted:** Mentioned the need for an option to go back from the public/private selection; found the current structure slightly rigid.
- **Improvement Suggested:** Ability to zoom in/out on the map, and expand beyond the sole use of filters in future versions.

Participant 2

- **Experience:** Described the process as “not that easy, not that hard.” Encountered some confusion with filtering options.
- **Positive Feedback:** Appreciated the multi-filter functionality and said it helped locate parking easily.
- **Issues Noted:** Was confused when certain filters seemed to produce repetitive or unclear results (e.g., public/private mixed with time).
- **Improvement Suggested:** No major suggestions, but highlighted confusion as a key usability point.

Participant 3

- **Experience:** Described the app as “great” and found it easy to book parking.
- **Positive Feedback:** Praised the booking functionality and the app's visual design.
- **Issues Noted:** Struggled to find the filter option initially due to color/design; once guided, navigation was smooth.
- **Improvement Suggested:** Make the filter button more prominent by adjusting the color scheme or size.

Participant 4

- **Experience:** Found the application clear and logical.
- **Positive Feedback:** Especially liked the clarity of filters and their ability to simplify booking; praised direction feature at the end.
- **Issues Noted:** No major issues noted; participant felt confident during the flow.

- **Improvement Suggested:** Include more visible options or wider selections for payments (e.g., Apple Pay).

Participant 5

- **Experience:** Described the experience as “pretty easy” and appreciated the simplicity of the interface.
- **Positive Feedback:** Highlighted the integration of price and EV charging options as highly useful; liked the payment integration.
- **Issues Noted:** None explicitly mentioned; participant navigated comfortably.
- **Improvement Suggested:** No strong suggestions generally satisfied.

Analysis of Test result

1. **Ease of Use:** 4 out of 5 users found the app intuitive and easy to use. One participant encountered minor confusion, which was quickly resolved.
2. **Filter Functionality:** Universally praised but initially hard to locate or interpret by some users. Suggestions revolved around enhancing visibility and simplifying filter combinations.
3. **Visual Design:** Received strong praise, especially for clarity and minimalism. However, some noted that **certain buttons (like filters)** blended too much with the background or weren’t immediately visible.
4. **Scenario Fit:** All users successfully completed the scenario (finding long-term, affordable parking near campus), validating the task alignment with real-world behavior.
5. **Missing Search Bar:** While not the main focus of this test, two participants mentioned the absence of a search bar as a limitation. This will be important to implement in the full version.
6. **Booking Confirmation & Navigation:** Users were able to complete the booking process and appreciated the ability to see directions afterward. One user emphasized that it reduced guesswork and saved time.

Findings and Recommendations

After analyzing all five participant transcripts from our usability testing, we identified key themes, usability successes, and pain points. Our findings are categorized below with specific user data and qualitative feedback to support each point.

Finding 1: Filter Feature Appreciated But Initially Confusing

Observation: 4 out of 5 participants expressed that the **filter feature** was very useful in narrowing down search results for parking based on time, price, and location. However, **3 participants** experienced hesitation or confusion locating or using it initially.

- Participant 1 said, “It was pretty straightforward,” but mentioned a desire for “an option to go back” and expressed mild confusion around public/private options.
- Participant 3 mentioned, “The color of the filter option wasn’t clear. I needed help to find it.”
- Participant 4 also echoed that “the filter button could be more visible.”

Recommendation:

In the next prototype:

- Improve the **visual prominence** of the filter button (larger size or contrasting color).
- Consider adding an **onboarding tooltip** or animation that highlights the filter option during first-time use.

Finding 2: Absence of Search Bar Led to Initial Confusion

Observation: Although participants were informed in advance that the search bar was non-functional in this prototype, **2 participants instinctively attempted to search manually** and expressed confusion when nothing happened.

- Participant 1 said, “It would be better if you didn’t have to use the filter alone.”
- Participant 5 asked, “Now what do I do?” upon encountering the disabled search bar.

Recommendation:

- Clearly **label prototype limitations** within the interface (e.g., greyed out buttons or "Coming Soon" labels).
- In final versions, **reintroduce the search bar** for broader user preference and accessibility.

Finding 3: Booking and Confirmation Was Smooth and Well-Received

Observation: All five participants **successfully booked a parking spot** using the prototype and appreciated the clarity of the process.

- Participant 3 said, “I liked the booking part most... The design is very good.”
- Participant 4 described the process as “really clear,” especially the direction function.
- Participant 5 said, “The interface with payment options was straightforward.”

Recommendation:

- Retain and slightly enhance the **confirmation and direction flow**, perhaps by adding animation or progress indicators.
- Add a **"Back" button** on the final confirmation page for better control.

Finding 4: Scenario Fit and Realism Appreciated

Observation: All participants quickly understood and related to the test scenario. The urgency of needing parking before class resonated well.

- Participant 2 noted that filters helped avoid “looking all over” and made the app’s purpose clear.
- Participant 4 said, “This is the problem many students face.”

Recommendation:

- Keep the scenario-based testing for future rounds.
- Embed a similar scenario or “Quick Start” option in the actual app to simulate real-life urgency.

Finding 5: UI Layout and Visual Feedback Need Minor Tweaks

Observation: Although the interface was praised for being clean and minimal, **2 participants struggled with button colors, icons, or layout logic.**

- Participant 3: “You can change the filter button color or make it more visible.”
- Participant 1: “I wanted a way to go back when I selected something wrong.”

Recommendation:

- Use **standard UI color conventions** (e.g., blue for primary actions, red for cancel).
- Provide **button press animations** or loading indicators.
- Add **navigation breadcrumbs** or “Back” buttons where applicable.

Finding 6: Confusion Between Public and Private Parking Options

Observation: Multiple participants expressed uncertainty when selecting between public and private parking during the filtering process. The terminology was unclear to some, and the distinction between the two options was not immediately obvious in the interface.

- Participant 1 hesitated when reaching this stage and mentioned, “I guess you want an option to go back from there, right?” suggesting difficulty in understanding the context.
- Participant 3 asked, “What’s the difference between public and private?” indicating the need for clarification.
- Participant 5 required an explanation from the facilitator to proceed.

Recommendation:

In the next version of the prototype:

- Provide **brief tooltips or definitions** next to public/private options (e.g., “Public: city-owned meters or lots; Private: business/residential spaces”).

- Use **icons or visuals** to distinguish between the two types.
- Consider adding a **“Learn More”** or **“What’s This?”** link to guide unfamiliar users during the filter selection process.

Summary of Key Improvements for Next Prototype

Area	Recommendation
Filter Button Visibility	Increase size, contrast, and possibly animate on first use
Search Functionality	Include a limited search feature or indicate its pending availability
Booking Flow	Add a “Back” option and visual indicators during confirmation
Visual Design	Adjust color scheme to ensure button contrast and icon clarity
Help/Guidance	Use tooltips or subtle instruction bubbles for new users

Conclusion

The usability testing of the *Park at Ease* prototype revealed that while users generally found the app straightforward and effective in helping them locate parking quickly, several areas require refinement to improve the overall user experience. Participants appreciated the filtering system and the clean interface, but some initially struggled to locate key features such as the filter button and desired clearer navigation cues. Minor adjustments to visual design, button feedback, and feature discoverability, especially regarding the search functionality, will be essential in the next prototype version. Overall, the app showed strong potential, with users successfully completing tasks and recognizing its value in real-world campus scenarios.